

Student Study Hours Analysis

Overview

This project uses R to explore the relationship between study hours and exam scores. A small dataset of five students is used to calculate average study hours, average exam scores, and the correlation between study time and exam performance.

Project Structure

- * `student-analysis.R` contains the R code used for the analysis.
- * `README.md` provides information about the project.
- * `script-documentation.Rmd` documents the R script.
- * `Reflection.md` contains the assignment reflection questions and answers.

Installation

To run this project, install R and RStudio or use Posit Cloud.

No external dataset is required because the data is created directly within the R script.

Dependencies

This project uses base R functions and does not require additional packages.

Example Usage

Open `student-analysis.R` and run the script.

The script:

1. Creates the student dataset.
2. Displays the data.
3. Calculates the average study hours.
4. Calculates the average exam score.
5. Calculates the correlation between study hours and exam scores.
6. Creates a scatter plot.

Example:

```
```r
average_study_hours <- mean(students$Study_Hours)
average_exam_score <- mean(students$Exam_Score)
correlation <- cor(students$Study_Hours, students$Exam_Score)
```
```

Results

The analysis calculates an average of 4 study hours and an average exam score of 72.8.

The correlation calculation shows the relationship between study hours and exam scores in the sample.

License

This project is created for educational purposes as part of the BDA400 Data Science Tools and Techniques course.

AI Assistance Disclosure

AI assistance was used to help plan and format the project documentation and to generate an initial version of the Markdown content.

The main prompts included:

- * "Explain what sections a good GitHub README for an R data analysis project should include."
- * "Revise the sections list so it's concise and uses Markdown headers and bullet formatting."
- * "Check the Markdown syntax for correctness and readability."

The AI-generated content was reviewed and edited to match the project requirements and the student's actual R script.